# GymBro · Architecture

Android app in Kotlin with MVVM (View → ViewModel → Repository → data sources) · own Flask REST API on Vercel (PaaS) · Firebase Cloud Firestore

## ANDROID APP · on the phone · Kotlin · minSdk 33

### DEVICE · sensors and system services

**Camera and photo picker**
camera app (TakePicture, FileProvider)
or the system photo picker
PictureChooser + ProfilePictures:
image processing (upright, square,
512 × 512 JPEG) → profile picture

**GPS + step counter**
Fused Location Provider, 1 fix / 2 s
step counter sensor (SensorManager)
RunTrackingService: foreground service
→ RunTracker → Run/Main ViewModels

**Microphone + speech recognizer**
AudioRecord: loudness only
SpeechRecognizer: laps said aloud
VoiceLapCounter → SwimSessionViewModel

**Network state**
ConnectivityManager callback:
back online → SyncManager.sync()

**Geocoder**
places searched in the route planner
→ RouteRepository

**Runtime permissions**
location, physical activity (steps),
microphone, notifications:
asked only when they are needed
camera: no permission needed
(the camera app takes the photo)

### UI LAYER · View

Activities and Fragments with XML layouts: they only draw the state and pass the user's actions to their ViewModel

| | | | |
|---|---|---|---|
| WelcomeActivity | MainActivity | HomeFragment | CalendarDrawer |
| Gym/Run/Swim fragments | RoutePlannerActivity | ChatActivity | PlanActivity |
| SocialActivity | UserProfileActivity | FollowListActivity | MessagesActivity |
| ConversationActivity | ProfileActivity | BottomNavBar | Google Maps SDK |

user actions → ViewModel functions

StateFlow<State> + one-shot events

### VIEWMODEL LAYER · one ViewModel per page (androidx.lifecycle)

read-only StateFlow state · one-shot events (Channel) · coroutines in viewModelScope · repositories injected by a Factory

| | | | |
|---|---|---|---|
| WelcomeViewModel | MainViewModel | HomeViewModel | GymSessionViewModel |
| RunSessionViewModel | SwimSessionViewModel | RoutePlannerViewModel | ChatViewModel |
| SocialViewModel | UserProfileViewModel | FollowListViewModel | MessagesViewModel |
| ConversationViewModel | PlanViewModel | ProfileViewModel | unit tests: fake repos |

calls the repository interfaces
(suspend functions, Flow)

**DOMAIN LAYER · use case**
SessionPlanChanges: gym session edits → plan

### DATA LAYER · Repository pattern (interface + Impl)

one instance each, created in AppContainer (manual dependency injection) · offline first: sync when back online

| | | | |
|---|---|---|---|
| UserRepository | AccountRepository | SessionRepository | PlanRepository |
| ProfileRepository | SocialRepository | MessageRepository | ChatRepository |
| RouteRepository | SyncManager | Notices → toasts | AppContainer (DI) |

Room DAO

read / write

suspend calls

**Room database (SQLite)**
AppDatabase, DAOs with Flow
training_sessions, plans,
posts, profiles, chat_messages
the pages show these data,
also offline

**SharedPreferences, files**
UserSession: login token, user,
changes not sent yet (offline)
ActiveSession, RunTracker,
SwimTracker: session in progress,
kept when the app is closed

**ApiClient (network)**
REST + JSON over HTTPS
HttpURLConnection, org.json
suspend, on Dispatchers.IO
Authorization: Bearer token
ApiException (0 = offline)

HTTPS · REST + JSON · Bearer token

## CLOUD

**Google Maps Platform**
map tiles for the Maps SDK of the app (key in local.properties)

**GitHub repository (this code)**
every push to main → the server is deployed again on Vercel

**Vercel · PaaS**
serverless functions, Hobby plan (free)
region fra1 (Frankfurt)

**Flask REST API (Python)**
`server/app.py`

- register, login, logout
- account, body data, appearance
- finished sessions, workout plans
- posts and personal bests
- search, follows, feed
- public and shared workout plans
- walking routes of the runs
- AI coach: chat and new plans

passwords: PBKDF2-SHA256 hashes
login tokens: random, saved hashed
service keys: only on the server

**Walking routes**
1. OpenRouteService
2. OSRM (FOSSGIS)
3. Google Routes API
the first one that answers

**AI coach (LLM)**
1. Groq: gpt-oss, Qwen
2. Google Gemini
plans as JSON (schema)
the first model that answers

Firestore client library (service account)

**Firebase Cloud Firestore · NoSQL database (Google Cloud)**
collections: users, posts, sessions, plans, follows, shares,
tokens, pictures, usernames, emails, counters
Spark plan (free), in Europe

**Secrets: never in git**
app: MAPS_API_KEY and API_BASE_URL in local.properties
server: environment variables on Vercel
(Firebase service account, OpenRouteService, Groq, Gemini)

Arrows: calls and data flow. AppContainer creates every repository once; the ViewModel factories inject them (in the unit tests the ViewModels receive fake repositories).

# GymBro · Navigation graph

The 43 screens and windows of the app (screenshots). Every arrow is labeled with the action that opens the next screen. Not drawn: Back ‹, Cancel and closing a window return to the previous screen. Dashed tags: the arrow continues at the screen with that number.

## ACCESS

App opened

logged in

not logged in

01 **Welcome**
WelcomeActivity

Already a member? Log in

03 **Log in**
window · WelcomeActivity

Log in

REGISTER

02 **Register**
window · WelcomeActivity

Register (account created)

04 **Personal information**
window · MainActivity
*asked once, after registering*

Save / Later

tap the picture

→ 28 Profile picture

## HOME

BOTTOM BAR · on the 4 main pages (and during a session): Home · Chat · Social · Profile

Home

05 **Home**
MainActivity · HomeFragment

START TRAINING

09 **Start training**
window · HomeFragment

calendar icon

06 **Calendar**
drawer · CalendarDrawer

tap a day with a dot

07 **Sessions of a day**
popup · CalendarDrawer

tap a session

08 **Session details**
window · SessionDetails
*the same window shows a post tapped in 20, 22 and 27*

tap a session

## TRAINING SESSIONS

Gym (no active plan, or a 1-day plan)

Gym (plan with 2+ days)

10 **Which day?**
window · GymSetup

choose a day

11 **Gym session**
GymSessionFragment

+ Add exercise
✎ edit

12 **Exercise**
window · ExerciseEditor

Stop ■

Run

Just run, without a goal

13 **Run goal**
window · RunSetup

Distance

36 **Goal distance**
window · RunSetup

Start run

15 **Run session**
RunSessionFragment

Stop ■

18 **Stop the session?**
window · SessionFragment

Save & publish or Save only

→ 05 Home

Route

14 **Route planner**
RoutePlannerActivity

Start run

Swim

16 **Swim setup**
window · SwimSetup

Start swim

17 **Swim session**
SwimSessionFragment

Stop ■

## AI COACH

Chat

19 **Coach chat**
ChatActivity

*✎ New chat: a new conversation*

## SOCIAL

Social

20 **Social feed**
SocialActivity

type in Search athletes

21 **Search athletes**
SocialActivity

tap the name of a post

tap an athlete

Messages icon

24 **Messages**
MessagesActivity

tap a conversation

25 **Conversation**
ConversationActivity

tap the name at the top

22 **User profile**
UserProfileActivity

Share a plan (you follow each other)

tap a public workout plan

Share a workout plan

tap a shared plan

43 **Share a workout plan**
window · PlanPicker

26 **Plan of another user**
PlanActivity

Add to my plans

→ 34 Workout plan

↓ Followers / Following
↑ tap an athlete

23 **Followers / following**
FollowListActivity

Followers / Following

## PROFILE AND WORKOUT PLANS

Profile

27 **Profile**
ProfileActivity

Menu ☰

29 **Menu**
drawer · ProfileActivity

Customize account

37 **Personal information**
menu · ProfileActivity

Personal information

30 **Customize account**
menu · ProfileActivity

Log out

→ 01 Welcome

tap the profile picture

28 **Profile picture**
window · PictureChooser
*Take a photo → camera app*
*Choose from gallery → photos*

Manage account

31 **Manage account**
menu · ProfileActivity

Modify account

38 **Modify account**
menu · ProfileActivity

+ New plan

33 **New workout plan**
window · ProfileActivity

Create the plan

tap a workout plan

34 **Workout plan**
PlanActivity

Ask the coach to change it

Plan updated see it

35 **Plan chat**
ChatActivity (about a plan)

Delete account

39 **Delete account**
window · ProfileActivity

Delete

→ 01 Welcome

A workout plan

A result added manually

+ Publish

32 **Publish**
window · ProfileActivity

A finished session

40 **Choose a session**
window · ProfileActivity

41 **Add a result**
window · ProfileActivity

42 **Publish a workout plan**
window · PlanPicker